A B C A L C U L U S

CHAPTER 5

# Analytical Applications of Differentiation

*AP-Style Practice Questions*

EASY MEDIUM HARD

Topics **5.1 – 5.12** AB

Name: _____ Period: _____

## PART I

*Topics 5.1 – 5.12*

## Multiple Choice Questions

*Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.*

Q1 EASY 5.2 MVT No Calculator

Let $f(x) = x^2 - 3x$ on $[0, 4]$. The value $c$ guaranteed by the MVT is

**(A)** $1$

**(B)** $2$

**(C)** $3$

**(D)** $\frac{3}{2}$

Q2 EASY 5.3 Critical Points No Calculator

The critical points of $f(x) = x^3 - 3x$ are

**(A)** $x = 0$ only

**(B)** $x = \pm 1$

**(C)** $x = \pm\sqrt{3}$

**(D)** none

Q3 EASY 5.4 Increasing / Decreasing No Calculator

$f(x) = x^3 - 6x^2 + 9x$ is decreasing on

**(A)** $(1, 3)$

**(B)** $(-\infty, 1)$

**(C)** $(3, \infty)$

**(D)** $(0, 3)$

Q4 MEDIUM 5.5 First Derivative Test No Calculator

If $f'$ changes from negative to positive at $x = c$, then $f$ has

**(A)** a local maximum at $x = c$ .

**(B)** a local minimum at $x = c$ .

**(C)** an inflection point at $x = c$ .

**(D)** no extremum at $x = c$ .

Q5 MEDIUM 5.6 Concavity No Calculator

$f(x) = x^4 - 6x^2$ is concave up on

**(A)** $(-\infty, -1) \cup (1, \infty)$

**(B)** $(-1, 1)$

**(C)** $(-\sqrt{3}, \sqrt{3})$

**(D)** everywhere

Q6 MEDIUM 5.7 Second Derivative Test No Calculator

For $f(x) = x^3 - 3x$, at $x = -1$ the function has

**(A)** a local maximum (by $f''(-1) < 0$ ).

**(B)** a local minimum (by $f''(-1) > 0$ ).

**(C)** an inflection point.

**(D)** neither max nor min.

Q7 HARD 5.8 Points of Inflection No Calculator

The graph of $y = x^5 - 5x^4$ has a point of inflection at

**(A)** $x = 0$ only

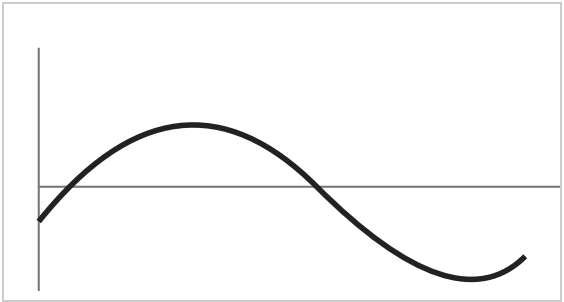
**(B)** $x = 3$ only

**(C)** $x = 0$ and $x = 3$

**(D)** $x = 0$ and $x = 4$

Q8 MEDIUM 5.4 Graph of $f'$ No Calculator

The graph of $f'$ is shown. On which interval is $f$ increasing and concave down?



**(A)** where $f' > 0$ and $f'' > 0$

**(B)** where $f' > 0$ and $f'' < 0$

**(C)** where $f' < 0$ and $f'' > 0$

**(D)** where $f' < 0$ and $f'' < 0$

Q9 MEDIUM 5.9 Candidates Test No Calculator

The absolute maximum of $f(x) = x^3 - 3x$ on $[-2, 2]$ is

**(A)** $-2$

**(B)** $0$

**(C)** $2$

**(D)** $4$

Q10 HARD 5.10 Optimization No Calculator

A rectangle with one side on the $x$-axis has its upper two vertices on the parabola $y = 12 - x^2$. The maximum area of such a rectangle is

**(A)** $16$

**(B)** $24$

**(C)** $32$

**(D)** $48$

Q11 MEDIUM 5.11 Implicit Optim No Calculator

The point on $y = \sqrt{x}$ closest to $(3, 0)$ has $x$-coordinate

**(A)** $\frac{3}{2}$

**(B)** $\frac{5}{2}$

**(C)** $1$

**(D)** $3$

Q12 EASY 5.12 Graph Sketching No Calculator

If $f'(x) > 0$ and $f''(x) < 0$ on $(a, b)$, then the graph of $f$ on $(a, b)$ is

**(A)** increasing and concave up

**(B)** increasing and concave down

**(C)** decreasing and concave up

**(D)** decreasing and concave down

Q13 HARD 5.9 Closed-Interval Test No Calculator

On $[0, 2]$, the absolute maximum of $f(x) = x\sqrt{2 - x}$ is

**(A)** $\sqrt{2}$

**(B)** $\dfrac{4\sqrt{6}}{9}$

**(C)** $2$

**(D)** $\dfrac{4}{3}$

Q14 MEDIUM 5.7 Concavity Inference No Calculator

The table gives values of a continuous function $f''$. The graph of $f$ must have an inflection point in which interval(s)?

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $f''$ | $-3$ | $-1$ | 2 | 1 | $-1$ |

**(A)** $(0, 1)$

**(B)** $(1, 2)$

**(C)** $(3, 4)$

**(D)** Both (B) and (C)

Q15 HARD 5.10 Optimization (Area) Calculator

A page must have a printed area of $96$ in², with $1$ in. margins on each side and $1.5$ in. margins top and bottom. The dimensions (width × height) that minimize total page area are closest to

**(A)** $9 \times 14$

**(B)** $10 \times 13$

**(C)** $11 \times 12$

**(D)** $12 \times 11$

Q16 MEDIUM 5.3 Graph of $f'$ No Calculator

If the graph of $f'$ crosses the $x$-axis at $x = 1$ (from + to −) and has a local min at $x = 3$, then $f$ has

**(A)** a local max at $x = 1$ and an inflection point at $x = 3$ .

**(B)** a local min at $x = 1$ and a local max at $x = 3$ .

**(C)** an inflection point at $x = 1$ and a local min at $x = 3$ .

**(D)** a local max at $x = 1$ and a local min at $x = 3$ .

Q17 MEDIUM 5.2 Rolle's Theorem No Calculator

Which hypothesis of Rolle's Theorem is violated for $f(x) = |x|$ on $[-1, 1]$?

**(A)** $f$ not continuous on $[-1, 1]$ .

**(B)** $f$ not differentiable on $(-1, 1)$ .

**(C)** $f(-1) \neq f(1)$ .

**(D)** None — Rolle's applies.

Q18 HARD 5.12 Sketch Reasoning No Calculator

Which of the following must be true if $f''(x) > 0$ for all $x$?

**(A)** $f$ is increasing.

**(B)** $f'$ is increasing.

**(C)** $f$ has a minimum.

**(D)** $f$ is linear.

**PART II** *Show All Work*

# Free-Response Questions

*Curve-sketching and optimization FRQs require sign charts for $f'$ and $f''$, clear tests (first / second derivative / closed-interval), and interval justifications.*

FRQ 1 EASY 5.3 – 5.7 Curve Analysis No Calculator

Let $f(x) = x^3 - 6x^2 + 9x + 2$.

**(a)** Find all critical points of $f$.

**(b)** Use a sign chart of $f'$ to classify each critical point as a local max, local min, or neither.

**(c)** Find the inflection point(s) of $f$ and the intervals of concavity.

FRQ 2 MEDIUM 5.9 Absolute Extrema No Calculator

Let $f(x) = \dfrac{x}{x^2+1}$ on $[0, 3]$.

**(a)** Find all critical points of $f$ on $[0, 3]$.

**(b)** Use the closed-interval method to find the absolute max and min of $f$ on $[0, 3]$.

**(c)** Find the intervals on which $f$ is concave up, and list any inflection points.

FRQ 3 MEDIUM 5.10 Optimization Calculator

A rectangular box with a square base and open top has volume $256$ in³. Material for the base costs $\$6$/in² and for the sides costs $\$2$/in².

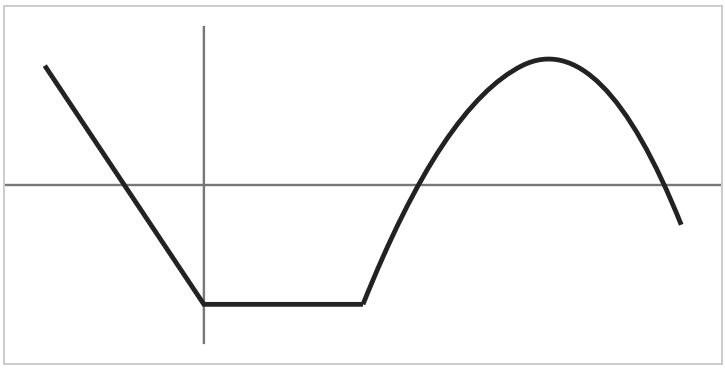
**(a)** If $x$ is the base edge and $h$ the height, write the cost $C(x)$ in terms of $x$ alone. Show the setup.

**(b)** Find the critical point(s) of $C$ on $(0, \infty)$ and justify that it gives a minimum using $C''$ or a sign chart.

**(c)** Report the minimum cost, rounded to the nearest cent, along with the dimensions that achieve it.

FRQ 4 HARD 5.4 / 5.7 Graph of $f'$ No Calculator

The graph of $f'$, the derivative of a function $f$, is shown on $[-2, 6]$. The graph consists of two line segments and a parabolic arc.



**(a)** Identify all $x$-values on $(-2, 6)$ at which $f$ has a local maximum. Justify.

**(b)** Find all $x$-values on $(-2, 6)$ at which $f$ has a point of inflection. Justify.

**(c)** Given that $f(-2) = 3$, determine whether $f(0)$ is greater than, less than, or equal to $f(-2)$. Justify using the graph of $f'$.

FRQ 5 HARD 5.10 Optimization & Geometry No Calculator

A farmer has $800$ meters of fencing and wants to enclose a rectangular field along a straight river (no fencing is needed along the river) and then divide it with one fence parallel to the river.

**(a)** Let $x$ be the length parallel to the river and $y$ be the length perpendicular to the river. Write an equation relating $x$ and $y$ from the fencing constraint.

**(b)** Express the enclosed area $A$ as a function of $x$ alone, stating the domain.

**(c)** Find the critical point of $A$ and use the second derivative test to justify that it gives a maximum.

**(d)** Report the optimal $x$, $y$, and maximum area.